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Characterizing Visually Objectionable and Non-Objectionable Medullated Alpaca Fibers

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ABSTRACT

It is recognized that visually objectionable fibers (VOFs) are undesirable in quality products because they appear lighter in dyed garments. Nevertheless, although a great deal of work has been done on mohair to characterize VOFs, this is not the case for alpaca fibers. Hence, this study was undertaken to characterize VOFs and visually non-objectionable fibers (VNOFs) in alpaca. Thus, a range of 52 alpaca fiber samples were obtained and visually separated into VOF and VNOF categories, and their respective cross-sectional, medullation, and average fiber diameter (AFD) characteristics were measured, analyzed, and characterized. The incidence of VOFs was found to be less than 2.5% by weight on average and was higher in adult alpacas than in younger ones ($p < .05$). Cross-sectional characteristics, degree of medullation, AFD, standard deviation of AFD, % of fibers $\geq 30 \mu\text{m}$, and spinning fineness for VOFs were all greater than those for VNOFs. The correlations between AFD and the fiber cross-sectional characteristics were higher for VNOFs than for VOFs, except for ellipticity. The correlations between the cross-sectional characteristics and the degree of medullation were slightly lower for VOF than for VNOFs, except for ellipticity.

摘要

人们认识到，视觉不良纤维（VOF）在优质产品中是不受欢迎的，因为它们在染色服装中看起来更轻。然而，尽管已经对马海毛进行了大量的研究来表征VOF，但羊驼纤维的情况并非如此。因此，本研究旨在表征羊驼的VOF和视觉无不良纤维（VNOF）。因此，获得了52个羊驼纤维样品，并将其视觉分离为VOF和VNOF；并测量、分析和表征了它们各自的横截面、髓质和平均纤维直径（AFD）特征。发现VOF的发病率平均低于2.5%（按重量计），成年羊驼的发病率高于年轻羊驼（ $p < .05$ ）。VOF的横截面特征、髓化程度、AFD、AFD标准偏差、 $\geq 30 \mu\text{m}$ 的纤维百分比和纺丝细度均大于VNOF。除椭圆度外，VNOF的AFD与纤维横截面特性之间的相关性高于VOF。除椭圆度外，VOF的横截面特征与髓质程度之间的相关性略低于VNOF。

KEYWORDS

Cross-sectional characteristics; fiber diameter; fiber quality; medullation; South American camelid; visually objectionable fibers

关键词

横截面特征; 纤维直径; 纤维质量; 髓质化; 南美骆驼科; 视觉上令人反感的纤维

Introduction

Alpaca, llama, camel, angora, mohair, and other animal fleeces comprise both unmedullated and medullated fibers, with the latter being classified as visually objectionable fibers (VOFs) and visually non-objectionable fibers (VNOFs), with most of the work in this regard having been done on mohair (Hunter, Smuts, and Botha 2013). VOFs are generally relatively coarse, highly medullated, and have a milky-white appearance when undyed, and also appear lighter in shade, than other fibers, when dyed (Balasingam 2005). Such fibers are generally highly undesirable in many good-quality products, particularly in mohair, where their presence and level lead to a considerable price penalty (B. A. McGregor and Butler 2004). Although most of the work in this regard has been done on mohair,

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Paucar-Sullca et al. (2024) and E. Quispe-Peña and Quispe-Bonilla (2024) also investigated the presence of such fibers in alpaca fleece samples, since the level of VOF also impacts on alpaca fiber quality, due to their potential adverse effect on prickle, and dyed appearance (E. Quispe et al. 2022). In practice, the degree of prickle depends upon the type and structure of the fabric and garment, the fiber diameter (coarseness), and the length of the protruding fibers (>0.2 mm) (Garnsworthy et al. 1988). For wool, G. Naylor (1992) found that the percentage of fibers coarser than $30\text{ }\mu\text{m}$ is a good predictor of prickle in knitted fabrics, while Frank et al. (2014) found the degree of prickle to be affected by the fibers stiffness, which, in the case of medullated fibers, is essentially influenced by the type and extent of the medulla, as well as by fiber diameter.

Although some researchers were of the opinion that reducing the level of fibers, which have a medulla to fiber diameter ratio above 60%, would reduce prickle, others suggested that the nature and degree of medullation, whether continuous, strong, discontinuous, or fragmented, also affect the degree of prickle (Paucar-Sullca et al. 2024). In mohair, it was found that, when using the traditional American Society of Testing Materials (ASTM) criterion of 0.6 (medulla to fiber diameter ratio), approximately half of the fibers so classified as visually objectionable (i.e. “kemp”) would in fact not be visually objectionable as assessed subjectively (i.e. “non-kemp”), while when using the 60% criterion, about 18% of the visually non-objectionable medullated fibers would in fact be subjectively assessed as being visually objectionable, as i.e. “kemp-type” (Hunter, Smuts, and Botha 2013).

As already stated, although a considerable amount of work has been carried out on characterizing medullation in mohair (Hunter, Smuts, and Botha 2013), this is not the case for alpaca, the knowledge about the nature and characteristics of objectionable alpaca fibers still being very superficial. In fact, with the present state of science, it is not possible to objectively define, measure, and quantify VOFs in alpaca, and therefore also associate them with the alpaca fiber quality and to improve them genetically on otherwise.

In light of the above, the research, reported here, was carried out with the following aims: a) to objectively measure various dimensional and other fiber properties such as type and degree of medullation, the cross-sectional characteristics, and the diameter of VOFs and VNOFs, respectively, and thereby to arrive at an objective method to define and differentiate or distinguish between the two types of alpaca fibers, and in the future to determine the incidence of VOFs, and b) to establish the correlations and interrelationships between the relevant characteristics.

Material and methods

Location and duration

The focus of this study has been on white Huacaya alpaca fibers, since they are such high-quality fibers where the medullation, particularly the objectionable medullated fibers, can be a major quality and commercial problem. Using random sampling, a fairly representative range of white Huacaya alpaca samples (37 Females and 25 males), and covering different age groups (Milk tooth: 15; Two teeth: 15; Four teeth: 10; and Full mouth: 12), were obtained from the Lachocc South American Camelid Research and Development Center (CIDCS-Lachocc), of the Universidad Nacional de Huancavelica–Perú. This center is located at an altitude of 4490 m above sea level and $12^{\circ}45'38''$ latitude south and $75^{\circ}19'59''$ longitude west. The annual ambient temperature can vary from -5°C to 18°C , while the mean annual rainfall is ≈ 750 mm.

Various medullation, fiber diameter, and cross-sectional characteristics of the fiber samples were measured at the Laboratory of Research and Technological Development Center Natural Fiber's Tech SAC, located in La Molina, Lima, Perú.

Fiber samples

Fiber samples, of approximately 5 g each, were taken from the midside of 52 alpaca animals, according to the procedure described by Aylan-Parker and McGregor (2002). Each sample was placed into a paper bag, and the relevant details of the sample, namely age group and gender of the animal, were recorded. After this, the fiber samples were taken to the above-mentioned Laboratory for further tests.

Selection of objectionable and non-objectionable fibers

Approximately, half (i.e. 2.5 g) was taken from each of the original 52 samples; this being referred to as the “parent sample.” From each such parent sample, securely held with a clip, dust, vegetable matter, and other contaminants were blown out by a blast of directed air from a fan. After this process, the sample (now minus the dust and vegetable matter) was weighed between 0.60 and 2.21 g, with the average being 1.67 g. An experienced laboratory technician – with more than 3 year working in alpaca – then visually identified, removed, and counted the clearly different (milky-white and very coarse) fibers (VOF) when viewed in air against a black velvet background, under standard laboratory lighting conditions, this being according to the procedures used for mohair fibers (Figure 1) by Hunter, Smuts, and Botha (2013). The VOFs and VNOFs were counted and weighed before they were placed in zip-lock bags labeled according to the animal's gender (male or female) and age group.

VOF and VNOF medullation characterization

Evaluation of the medullation type of each fiber entailed measuring the profiles of the whole VOF, using a computerized projection microscope, called Medullometer (M. Quispe-Bonilla and Quispe-Peña 2018). This instrument is fitted with software that enables the fiber and medulla diameters to be measured and recorded according to the IWTO standard test methodology (IWTO 2015a). From each of the 52 parent fiber samples (without contaminants), 30 VOFs were randomly selected and their medullation type was analyzed at five different segments (sections) in each case (each 0.2–0.4 mm in length) along the length of the fiber. In addition, for each of the five fiber sections, the fiber and medulla diameters (AFD and AMD, respectively) were measured at two different positions. The medullation was classified into five different types as follows: non-medullated, fragmented, discontinuous, continuous, and strongly medullated (Figure 2), according to Wildman (1954) and Tridico (2009). Hence, the experimental design involved 52 samples $\times$ 30 fibers/sample $\times$ 5 observations/fiber, thereby producing a total of some 15,600 results. In a few isolated cases, not all the planned measurements could be carried out, with the actual total number of results being 15,582.

The medullation types observed in the five sections of each VOF were classified as follows (Figure 3):

- Two (02) categories: Monotype or heterotype (Hatcher 2002; Maqalika 2020). This description was applied when, in the five sections of the objectionable fibers, either only one (monotype) or more than one (heterotype) type of medullation was observed.

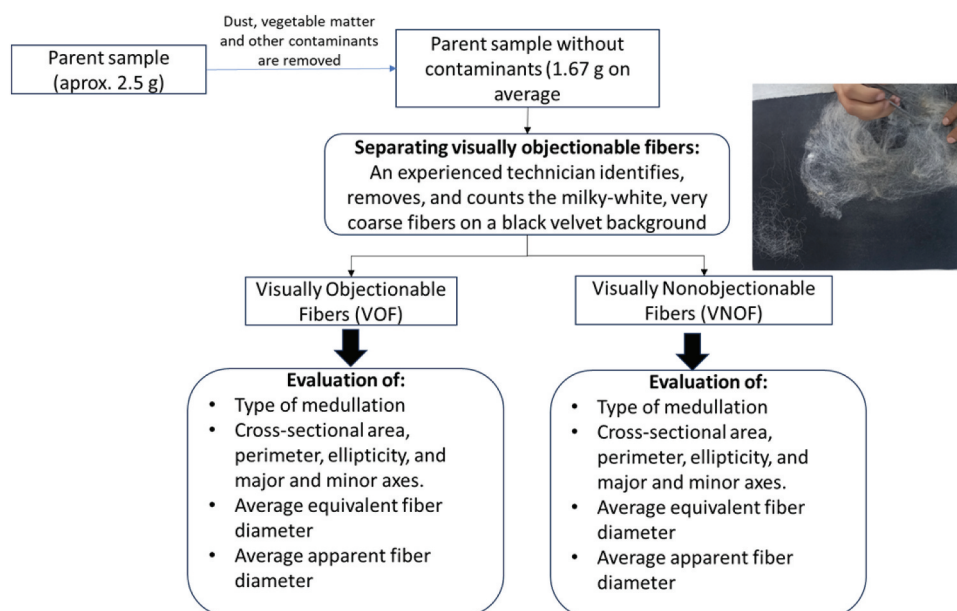


Figure 1. Identification, separation, and evaluation of the different types of medullated fibers.

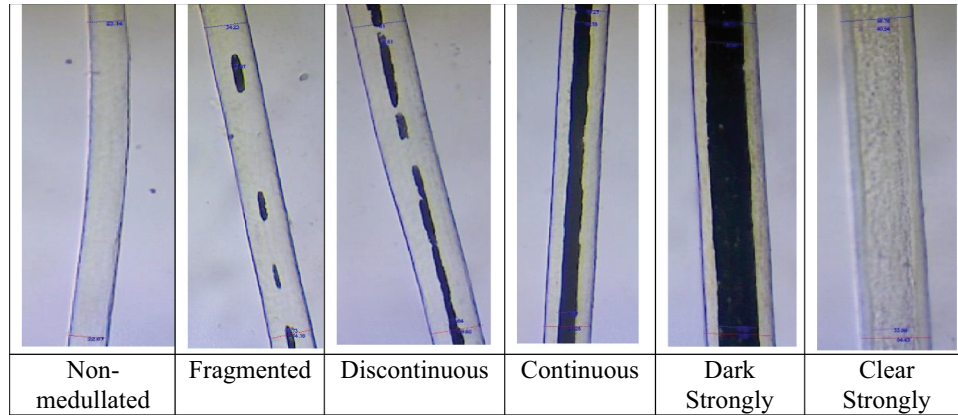


Figure 2. Images of the different types of medullated alpaca fibers as classified in this research.

- Four (04) classes: When one, two, three or four different medullation types were observed in the five sections of the VOF, they were termed as follows: one, two, three, or four medullation classes, respectively. Fibers with five different medullation types were not found.
- Four (04) predominance: This classification was used when, in the five sections, one specific medullation type, i.e. either, continuous, discontinuous, fragmented, or strongly medullated, predominated. There were no fibers with a predominance of “no medullation.”

The VOF and VNOF were also evaluated with the Fiber Med instrument (Model: Maxcorp; Version: V.2.0) according to the methodology proposed by M. D. Quispe-Bonilla et al. (2023), which provided the medullation characteristics listed in Table 1. In addition, the average fiber diameter (AFD) and its variation, for both VOF and VNOF, were measured with the Fiber EC instrument (Model: Small; Version: V.3.1), according to the Peruvian Technical Standard 131-400:2024 for Alpaca Fiber (INACAL 2024) and M. D. Quispe et al. (2017). The Fiber Med and Fiber EC were previously calibrated using eight standard wool tops with known measurements, according the procedure indicated by M. D. Quispe-Bonilla et al. (2023) and INACAL (2024), respectively. The various properties measured are listed in Table 1.

The fiber cross-sectional properties (Table 2) were measured with the Fiber Mult instrument (Model: NFT; Version 3.0) previously calibrated using a film calibrator that has divisions of 0.1 mm (NFT 2024), and using a novel procedure which involved preparing cross-sections of approx. 1.5 g fibers by means of Hardy’s modified microtome and razor blade as follows: A parallel fiber array of the specimen was clamped securely in the clamps of the modified Hardy’s microtome. Thereafter, fiber beards, projecting from both two sides of the clamps of the microtome, were cut with a sharp razor blade, and the cross-sectional characteristics of the remaining fibers were measured on the Fiber Mult (Figure 4).

In each cases, some 8 to 12 images were captured, with each containing between about 80 and 250 fiber cross-sectional images (Figure 5). The Fiber Mult automatically defines the perimeter of each fiber in each image and determines the different cross-sectional characteristics.

The average values of the characteristics measured, as described above, are given in Table 2. The equivalent fiber diameter (AFD_e) was obtained according to Kicińska-Jakubowska et al. (2024), as in Equation (1).

$$AFD_e = \sqrt{\frac{4 * Area}{\pi}} = \frac{Perimeter}{\pi} \quad (1)$$

were *Area* and *Perimeter* represent the average cross-sectional area (μm^2) and perimeter (μm), respectively. Furthermore, the apparent average fiber diameter (AFD_a) was derived from the means of the major and minor axes (Garat et al. 2018), as detailed in Equation (2).

$$AFD_a = \frac{\pi}{4} (x_{major} * x_{minor}) \quad (2)$$

were *x_{major}* and *x_{minor}* represent the major and minor axes, respectively, expressed as μm .

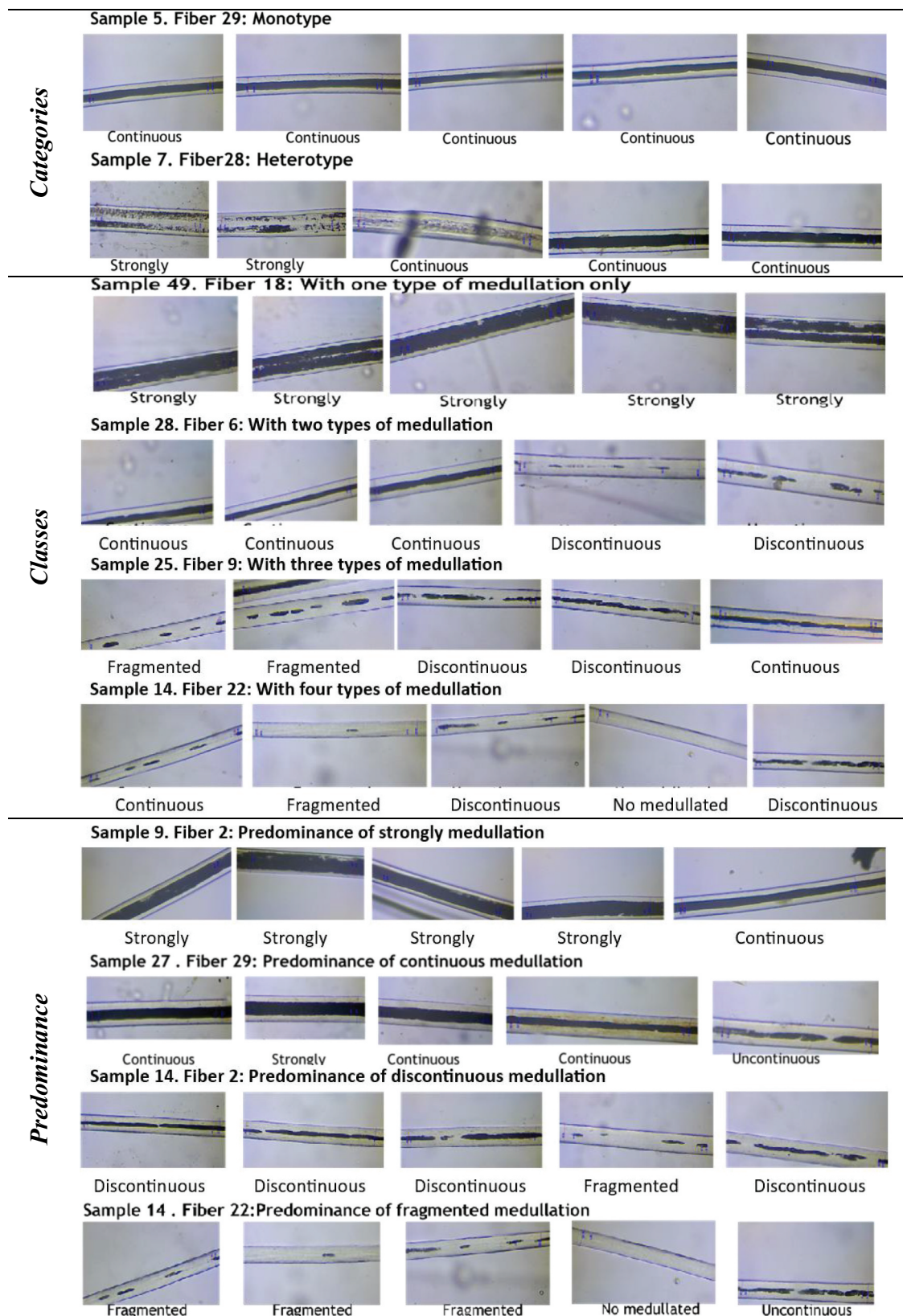


Figure 3. Categories, classes, and predominance of visually objectionable alpaca fibers.

Table 1. Key to abbreviations used in text and tables.

Abbreviation	Description and measure indicator
<i>Parameters determined by Fiber Med device</i>	
p_Med	Total medullation of fibers (%)
SE_p_Med	Standard error of p_Med (%)
p_NoMed	Non-medullated (%)
p_Frag_Med	Fragmented medullation (%)
p_Disc_Med	Discontinuous medullation (%)
p_Cont_Med	Continuous medullation (%)
p_Str_Med	Strongly medullated (%)
AFD _{FM}	Average fiber diameter (μm)
SD_AFD _{FM}	Standard deviation of AFD (μm)
AFD_NoMed	AFD of Non-medullated fibers (μm)
AFD_Frag_Med	AFD of Fragmented medullation fibers (μm)
AFD_Disc_Med	AFD of Discontinuous medullation fibers (μm)
AFD_Cont_Med	AFD of Continuous medullation fibers (μm)
AFD_Str_Med	AFD of Strongly medullated fibers (μm)
<i>Parameters determined by Fiber EC device</i>	
AFD _{FEC}	Average fiber diameter (μm)
SD_AFD _{FEC}	Standard deviation of AFD (μm)
CV_AFD _{FEC}	Coefficient of variation of AFD (%)
Fibers < 30 μm	Percentage of fibers < 30 μm (%)
Fibers ≥ 30 μm	Percentage of fibers ≥ 30 μm (%)
Spi_Fin	Spinning fineness (μm)
<i>Parameters determined by Fiber Mult device and procedure</i>	
Area	Average cross-section area (μm ²)
Perimeter	Average cross-section perimeter or contour (μm)
xminor	Average cross-section Minor axis (μm)
xmajor	Average cross-section Major axis (μm)
Ellipticity	Ratio of Major axis/Minor axis
AFD_e	Equivalent AFD (μm)
AFD_a	Apparent AFD (μm)

Table 2. Means ± standard error, and median (in bold) of VOF and VNOF, and number of VOF per gram according to animal age group and gender.

Factors	VOF (%)	VNOF (%)	Number of VOF/g (n)
<i>Age group</i>	[0.0164]	[0.0164]	[0.0650]
• Young	1.87 ± 0.18 1.58^a	98.13 ± 0.18 98.42^b	193.26 ± 13.63 193.00^a
• Adult	2.52 ± 0.24 2.11^b	97.48 ± 0.24 97.89^a	155.36 ± 14.91 126.67^a
<i>Gender</i>	[0.1985]	[0.1985]	[0.3229]
• Male	1.83 ± 0.15 1.72^a	98.17 ± 0.15 98.28^a	170.18 ± 16.31 148.11^a
• Female	2.32 ± 0.23 2.00^a	97.68 ± 0.23 98.00^a	185.94 ± 13.65 196.77^a
Total	2.11 ± 0.15 1.89	97.89 ± 0.15 98.11	179.22 ± 10.43 165.68

In brackets are the p-values obtained using the Kruskal–Wallis test. Medians with different superscripts in a column indicate significant differences to the Dunn test ($p < 0.05$) for each factor. VOF, visually objectionable fibers; VNOF, visually non-objectionable fibers.

Statistical analysis

Descriptive statistical analysis was used to characterize both the VOF and VNOF. To analyze the effect of the animal gender and age group on the percentage of VOF and VNOF, and on the level of VOF per gram, based on the mass of the parent sample (i.e. without dust and vegetable contamination), the Kruskal–Wallis and Dunn tests were carried out, since the dependent variables did not have a normal distribution and homoscedasticity. The Kruskal–Wallis test uses ranks in one-criterion variance analysis (Kruskal and Wallis 1952), and Dunn multiple comparison uses rank sums (Dunn 1964)

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(N+1) \quad (3)$$

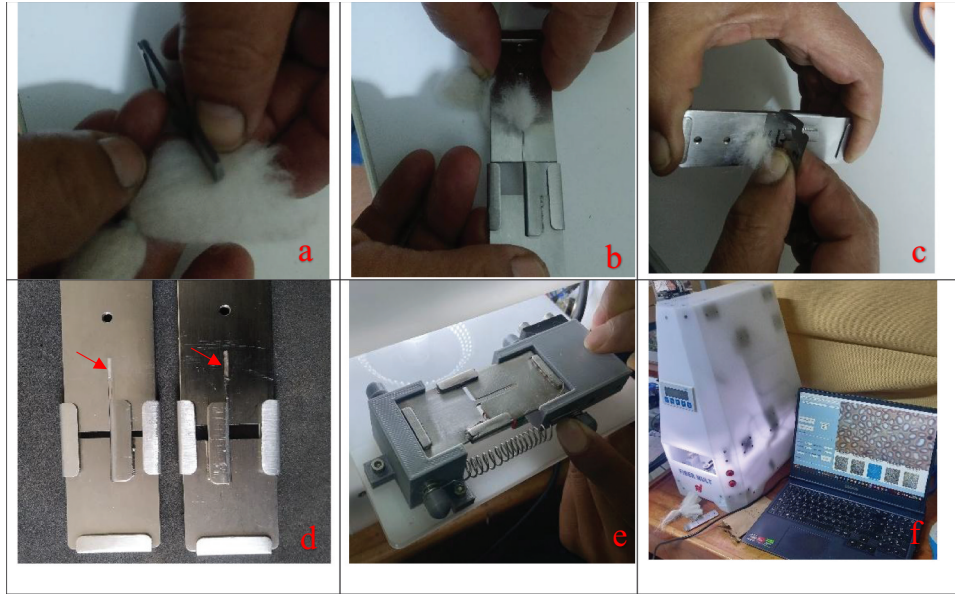


Figure 4. Steps involved in evaluating fiber cross-sectional characteristics using fiber Mult: a) placing the fiber sample in the modified Hardy microtome; b) clamping fibers; c) cutting the protruding fibers; d) fiber sample clamped in the microtome; e) placing the microtome with fibers in the fiber Mult holder; f) fiber Mult unit and computer analyzing cross-sectional characteristics of the fibers.

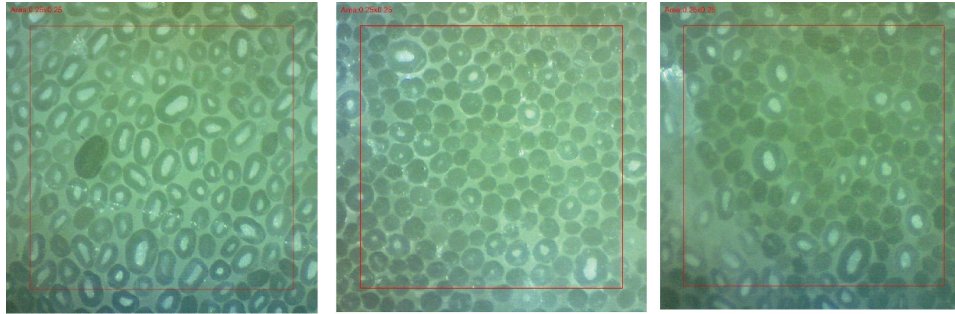


Figure 5. Cross-sectional images of alpaca fibers, captured by fiber Mult.

where: H : Kruskal–Wallis’s statistic, N : Total number of observations, R_i : Sum of ranks in the i -th age group or gender, n_i : Number of observations in the i -th age group or gender, and K : Number of groups (only two groups).

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \left(\frac{R_i^2}{n_i} \right) - 3(N+1) \quad (4)$$

where: z : Dunn’s statistic, N : Total number of observations, $\bar{R}_i$: Mean of the ranks of the i -th age group or gender. $\bar{R}_j$: Mean of the rank of the j -th group (age group or gender), n_i : Number of observations in the i -th age group or gender, n_j : Number of observations in the j -th age group or gender.

For the determination of the variance components of the average fiber and medulla diameters (obtained with the computerized projection microscope), a nested linear model was used, with the sample, fiber/sample, section/fiber/sample, and residuals being considered as randomized effects, according to Equation (5).

$$y_{ijk} = \mu + sam_i + fib_j(sam_i) + sec_k(fib_j(sam_i)) + e_{ijk} \quad (5)$$

were y_{ijk} is the average fiber and medulla diameter; μ is the global mean; sam_i is the i -th effect of the sample; $fib_j(sam_i)$ is the nested effect of the j -th fiber within the i -th sample; $sec_k(fib_j(sam_i))$ is the nested effect of the k -th section within the j -th fiber, within the i -th sample; and e_{ijk} is the residual error.

To analyze the categories, classes, and predominance of medullation type in VOFs, bar charts and tables, with both absolute (count) and relative (%) values, as well as Spearman correlations and tables, as appropriate, were used. For the different statistical analysis, free software rStudio was used (R Core Team 2024)

Results and discussion

Incidence of VOF and VNOF, and characteristics of VOF

The mean, standard errors, and medians of VOF and VNOF percentage by weight, and the number of VOFs by age group and sex, are shown in Table 2. From this table, it is apparent that only the age group had an effect on the VOF and VNOF expressed as a percentage by weight ($p < .05$), but not on the number of VOF/g, with animal gender not affecting any of these parameters. Thus, although the percentage of VOF by weight appears to be low (less than 2.2%), the actual number of VOF/g in alpaca fiber appears high (≈ 179 fibers/g). Paucar-Sullca et al. (2024) found an average VOF level of 1.93% by weight and 94 VOF/g by number, which are slightly lower than the levels observed here. This suggests that there is generally a low incidence of VOF in alpaca, although the number will vary depending on the source of the sample.

As can be seen from Table 2, the young animals had a lower percentage of VOF by weight than the adult animals, but had a higher percentage by number of VNOF, most likely due to the fact that the young animals had a lower AFD than the adult animals (Lupton, McColl, and Stobart 2006; Montes et al. 2008; E. C. Quispe et al. 2009). If we consider a length of 10 cm for a fiber sample with a density = 1.309 g/cm^3 (Czaplicki 2012), when the AFD of the alpaca fiber increases from 21 to $24 \mu\text{m}$, the weight (weight = density $\times$ volume) increases by more than 30%; thus, 100 fibers with an AFD = $24 \mu\text{m}$ would weigh more than 130 fibers with an AFD = $21 \mu\text{m}$.

For alpaca, there is no information about the relationship between VNOF and VOF. Hunter, Smuts, and Botha (2013) found that for mohair, with high and medium medullation – classification according to IWTO-57-98 (IWTO 2015b), there were between 320 and 475 VOF/g, which is greater than the present levels found here for alpaca fibers (179 VOF/g). This difference could be due to the fiber type (mohair vs alpaca) as well as the fact that the alpaca fibers considered here were finer (AFD = $23.4 \mu\text{m}$) than the mohair (AFD = $33.7 \mu\text{m}$). M. Quispe-Bonilla et al. (2022) found that a higher AFD was associated with a higher incidence of continuous and strongly medullated fibers, with such fibers generally being classified as VOF.

VOFs in wool and mohair generally have an opacity greater than 94%, as measured by an Optical Fiber Diameter Analyzer (OFDA). For alpaca fibers, however, the OFDA underestimates the number of objectionable fibers (E. C. Quispe et al. 2023), and this is most likely also the case for mohair (Lupton and Pfeiffer 1998). Hence, it may not be feasible to compare OFDA-based VOF levels for different animal species.

Table 3 presents the variance components of the average fiber and medulla diameters of VOFs, from which it is apparent that the fiber and medulla diameter variations between samples and between section/fiber/samples contributed more than 80% toward the total variation of the average fiber and medulla diameters, while the between fibers (within sample) variation contributed only 6.80% and 6.70% toward the variation in the average fiber and medulla diameters, respectively.

Table 3. Variance components of average fiber and medulla diameters of VOFs.

Variance components	Average Fiber Diameter		Average Medulla Diameter ¹	
	Absolute (μm^2)	Relative (%)	Absolute (μm^2)	Relative (%)
Between samples	30.80	40.40	33.282	44.40
Between fibers/samples	5.20	6.80	5.027	6.70
Between section/fiber/samples	33.40	43.80	29.795	39.80
Residual	6.80	8.90	6.775	9.00

¹Obtained with computerized projection microscope. Sample, fiber/sample, sections/fiber/sample, and residual are considered as components.

Various workers have investigated the variance components of AFD between fleeces and body areas/fleeces (Aylan-Parker and McGregor 2002; B. A. McGregor, Ramos, and Quispe Peña 2012; Príncipe 2025; Radzik-Rant, Wielechowska, and Rant 2021) as well as that between and within fleeces (Manso 2011; Príncipe 2025; E. Quispe et al. 2024), with the finding that the variation between fleeces represented between 37% and 90% of the total variation, with the residual representing between 40% and 60% of the total. Whereas along-fiber diameter variation has been investigated for wool and mohair, very little research has been carried out on alpaca fibers. Bustinza (2001), taking measurements every centimeter along fibers of 18-month-old alpaca animals, found a variation along the fibers of between 22.6 and 29.6 μm , although the relative contribution of this to the total variation in fiber diameter was not determined. Hence, it is very difficult to draw valid conclusions concerning variations along alpaca fibers.

For Merino wool, it was found that the diameter variation along the fiber normally represents only 16% of the total variation, whereas when the animals are subjected to stress (reduced nutrition of water and feed during lactation or pregnancy), it can increase to 43%, normally; however, the between fiber within staple variation is the major source (64%) of the total variation in fiber diameter. Furthermore, the body area within fleeces contributed only between 3% and 4% toward the total variation, while the variation between fleece and along fiber within the staple each represented 16%, of the total variation (Hansford 1992). On the other hand, E. C. Quispe-Peña et al. (2014) found that for vicuña, the variance between fleeces represented less than 30% of the total variation in AFD.

Our results reflect a high variation in AFD along VOF (greater than 40% corresponding to more than 33.4 μm^2), which may be due to the fact that alpacas are raised under extensive conditions in high Andean areas, and there they are at the mercy of food availability, which is a function of rainfall, there being two distinct periods: rainy and dry, with forage availability and quality much poorer during the dry period (Flores 1991; Siguayro 2008), resulting in a decrease in AFD during this period. Bustinza (2001) found that the variability in diameter along the fiber increased in proportion to the diameter, so that finer fibers were generally more uniform, while Franco et al. (2008) found that, in terms of diameter and length, the coarser fibers were affected more by low nutrition than finer fibers. Russel and Redden (1997) found that, for alpaca, a lower nutritional level led to a lower fiber growth rate, decreased fiber diameter, and a lower length/diameter ratio.

The results for the different types of medullation in VOF are captured in Table 4 from which it is apparent that almost 58% of VOF fell into the continuous medullation (Cont_Med), category with strongly medullated fibers (Str_Med) representing only 1.28%. Heterotype VOF contained relatively high levels of Cont_Med and Str_Med sections (24.76%), and also relatively high levels of Cont_Med and discontinuous medullation (Disc_Med) sections (Cont_Disc, 11.42%). Thus, approximately 85% of VOF are fibers with Cont_Med or Str_Med. Nevertheless, heterotype VOF, with different types of medullation sections, have lower levels. There were, however, no VOF with only fragmented medullated (Frag_Med) or non-medullated (NoMed) fibers.

It is known that as AFD increases, the medulla tends to become continuous along the entire fiber length, also becoming increasingly wider (Olarte et al. 2023; Pinares et al. 2019; Villarroel 1959), which agrees with our present results, where coarse VOF mainly contained Cont_Med and Str_Med sections.

Table 4. Number and percentage of VOFs according to medullation type ($n = 1559$ fibers).

Medullation category ¹	n ²	%	Medullation category ¹	n ²	%
<i>Monotypes</i>			Cont_Frag	18	1.15
Disc	2	0.13	Disc_Frag	5	0.32
Cont	904	57.99	Frag_Disc	2	0.13
Str	20	1.28	Cont_Str_Disc	3	0.19
<i>Heterotypes</i>			Cont_Disc_Frag	9	0.58
Cont_Disc ³	178	11.42	Frag_Disc_Cont	4	0.26
Str_Cont	119	7.63	Disc_Cont_Frag	4	0.26
Disc_Cont	8	0.51	Cont_Disc_Str	5	0.32
Cont_Str	267	17.13	Cont_Frag_Disc	4	0.26
Cont_NoMed	2	0.13	Str_Cont_Disc	1	0.06
Cont_Str_Frag	3	0.19	Cont_Disc_Frag_Str	1 ²	0.06

¹Medullation types: Disc, discontinuous; Cont, continuous; Str, strong; NoMed, non-medullated; Frag, fragmented. The order indicates the predominance from highest to lowest.

²Number of fibers.

Our results also agree with those of Pinares, Quispe, and Abd Manan (2024), namely that Con_Med and Str_Med fibers generally represent VOFs, with a high percentage of these fibers negatively affecting the appearance, processing, and textile performance of such Huacaya and Suri alpaca fibers. Hence, the reduction of VOF in alpaca, by selection and other breeding practices, is crucial for achieving the desired quality improvements. Although information about the variability of medullation type along the alpaca fiber, is relatively scarce, Villarroel (1959) showed that finer fibers are usually more uniform than coarser fibers.

There is, currently, some confusion between “strongly medullated fibers,” “objectionable medullated fibers,” and VOF. For alpaca, the term “strongly medullated fibers” was used by Pinares et al. (2019), based on the classification of Wildman (1954), while Tridico (2009) referred to fibers with unbroken (wide) lattices, where the mounting medium sometimes also entered into the medulla. According to the ASTM (1989), VOFs (kemp-type fibers) are classified as having medulla diameters 60% or greater than that of the fiber diameter. Thus, the term “strongly medullated fibers” was subsequently used in several studies involving alpaca fibers (Cruz et al. 2024; Olarte et al. 2023; Pinares, Quispe, and Abd Manan 2024; Pinares et al. 2019; E. Quispe et al. 2022). On the other hand, however, the term “objectionable fiber” can also be used for fibers having a medulla to fiber diameter ratio of 0.55 or more, when testing mohair or wool in benzyl alcohol (Smuts and Hunter 1987), or when more than 0.60, according to ASTM (1989). Balasingam (2005), on the other hand, suggested that objectionable medullated fibers be defined as fibers that are generally chalky-white in appearance, even after dyeing, and may be classified as kemp fibers, highly medullated fibers, and flat fibers.

The VOF term in mohair was defined by Hunter, Smuts, and Botha (2013) with respect to those clearly different (milky-white) fibers when viewed in air against a black velvet background under standard laboratory lighting conditions; thus, they can be visually identified, counted, and removed by an experienced laboratory technician. According to these researchers, a true kemp fiber is shed by goats at least annually and has a characteristic bulbous root and tapered tip. Hence, a VOF can be either a kemp or a non-kemp fiber; nevertheless, they also found large differences between different laboratories and/or technicians in the levels of VOF determined on the same samples. Additionally, Paucar-Sullca et al. (2024), during a preliminary evaluation of alpaca fiber medullation, were able to identify VOF.

In the present study, it has been found that aprox. 60% of VOF are monotypes, meaning that only one medullation type is present throughout the fiber, while in the rest (aprox. 40%) of the VOFs, i.e. heterotypes, different medullation types were present. Additionally, it was also found that slightly more than 94.6% of such heterotype VOFs contain two medullation types, while 5.2% have three medullation types. VOFs that contain four medullation types are very rare (0.2%), while there are no VOFs with five medullation types (Figure 6).

When the incidence of the predominant medullation types was evaluated, it was found that VOFs with Cont_Med predominate in almost 90% of the cases (alone or together with other medullation types), with Str_Med predominating in 9%. VOFs, in which Frag_Med and Disc_Med predominate, occur in less than 1.5% (see Table 4 and Figure 6).

Our results for the percentages of VOF with Cont_Med (57.99%) are similar to those (57.46%) obtained by Paucar-Sullca et al. (2024), although their results for the percentages of VOF with Frag_Med, Disc_Med, and Str_Med (1.05%, 4.68%, and 36.82%, respectively) are very different from the results obtained in this study. These differences may be due to the different method they used for determining the fiber medullation types, namely, in a single region (section), whereas we considered 20 types of medullation based upon the evaluation of five regions (sections) along the fiber.

The results obtained here, on the different medullation types, illustrate the heterogeneity of alpaca VOFs in terms of their medullation characteristics. VOFs are considered to be undesirable fibers in terms of the quality of mohair (Hatcher 2002; Hunter, Smuts, and Botha 2013; Smuts and Hunter 1987), cashmere (B. McGregor 2012), and wool (Balasingam 2005; Fleet, Mahar, and Turk 2002; A. L. Sánchez et al. 2016), and the same could also be applied to alpaca fibers. Therefore, for the purposes of the genetic improvement of alpaca fleeces, if the objective is to improve the quality of the fibers, before considering decreasing or eliminating the medullation or certain medullation types (fibers with Cont_Med and/or Str_Med) to reduce or eliminate prickles, it is rather important to consider eliminating at least or decreasing VOF levels. Thus, the VOF percentage could be used as a selection criterion.

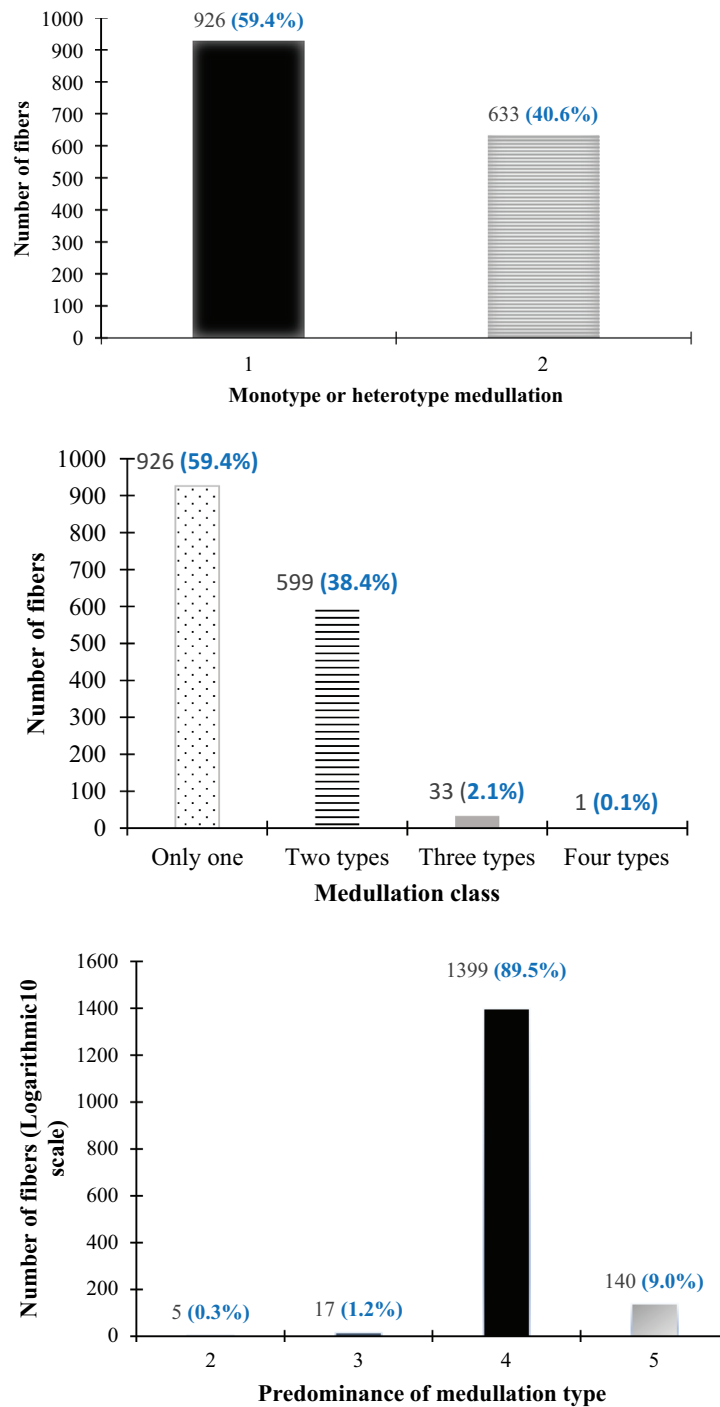


Figure 6. Incidence of monotype or heterotype, class and predominance of medullation types, of the visually objectionable alpaca fibers (in blue is expressed as percentage).

Nevertheless, this needs to be supported by additional information regarding the relevant genetic parameters and the effects of intrinsic and extrinsic factors which affect the incidence of VOF in alpaca fleeces.

The heterogeneity of the medullation types found here in the VOF (with a predominance of Cont_Med and Str_Med about 98%) disagrees to some extent with the findings of Hunter, Smuts, and Botha (2013) for mohair. They found only Cont_Med ($\approx 65\%$), lattice ($\approx 30\%$), or flat ($\approx 5\%$) medullation types in the VOF. This difference could be related to the type of animal (i.e. alpacas or goats), and also the inherent differences in alpaca fibers. For example, in alpaca fibers, we found no flat medullated fibers among the VOF (Figure 7).

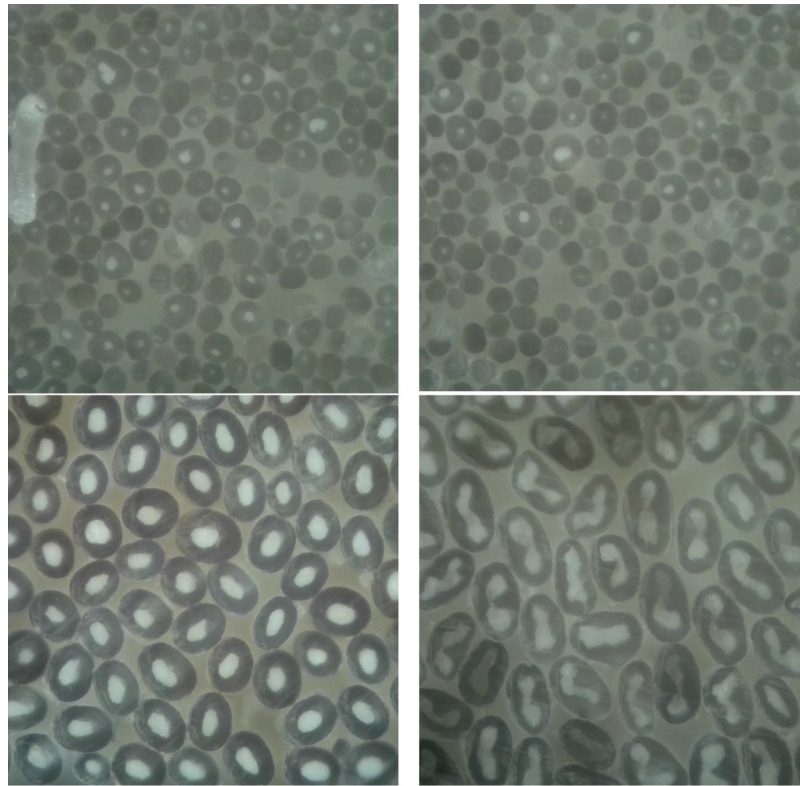


Figure 7. Cross-section images of visually non-objectionable (top) and objectionable alpaca fibers (bottom), captured by fiber Mult device. The difference between them is clear.

Cross-sectional characteristics

The cross-sectional results for the VOF and VNOF are shown in Table 5, from which it can be seen that the average cross-sectional area of VOFs is more than three times that of VNOFs (871.24 and $286.74 \mu\text{m}^2$, respectively), while the other VOF characteristics, such as perimeter, minor and major axes, AFD_e, and AFD_a, are close to 1.7 times greater than those of the VNOFs, although the ellipticity of the VOF is only 1.13 times greater than that of VNOF. It is also apparent that the average and median values of the various characteristics are very close, which indicates that these variables could have a normal distribution.

Our results on the cross-sectional area, perimeter, AFD_e, and AFD_a of white VNOF and VOF are unique for alpaca fibers, whereas a great deal of similar results are available for wool. Allain and Thébault (2000) found that Angora rabbit down fibers with an AFD of $15.4 \mu\text{m}$ had an average cross-sectional area of

Table 5. Summary statistics for cross-sectional characteristics of non-objectionable and objectionable alpaca fibers.

Variables	N	Mean	Standard deviation	Median	Minimum	Maximum	Standard error
<i>Non-objectionable fibers</i>							
Area (μm^2)	91,548	286.74	107.36	268.97	39.73	604.63	0.35
Perimeter (μm)	91,548	60.64	11.39	59.69	23.62	101.49	0.04
Minor axis (μm)	91,548	17.25	3.29	17.04	5.99	28.46	0.01
Major axis (μm)	91,548	21.51	4.33	21.06	8.21	44.06	0.01
Ellipticity	91,548	1.25	0.16	1.23	1.00	3.22	0.00
AFD_e (μm)	91,548	18.78	3.52	18.51	7.11	27.75	0.01
AFD_a (μm)	91,548	19.38	3.63	19.09	7.88	31.82	0.01
<i>Objectionable fibers</i>							
Area (μm^2)	34,241	871.24	305.66	796.98	490.95	4733.14	1.65
Perimeter (μm)	34,241	107.02	18.66	103.36	79.38	265.50	0.10
Minor axis (μm)	34,241	28.33	4.57	27.65	17.97	58.39	0.02
Major axis (μm)	34,241	39.67	8.04	38.11	25.78	108.71	0.04
Ellipticity	34,241	1.41	0.20	1.37	1.00	2.83	0.00
AFD_e (μm)	34,241	32.85	5.49	31.86	25.00	77.63	0.03
AFD_a (μm)	34,241	34.00	5.85	32.88	25.28	83.55	0.03

AFD_e, average equivalent fiber diameter; AFD_a, average apparent fiber diameter.

167 μm^2 . The same authors (1996) found the cross-sectional area, perimeter, and major and minor axes of mohair to be 796.5 μm^2 , 134.2 μm , 37.7 μm , and 29.5 μm , respectively, while for Angora rabbit fibres, the values were 180.8 μm^2 , 47.0 μm , 18.7 μm , and 13.6 μm , respectively. Hence, in terms of the AFD of the fibers tested here, the cross-sectional area results of the VOF and VNOF are very similar to those found by Allain and Thébault (1996) for mohair with respect to VNOF, since their AFDs are very similar.

According to the ellipticity (Table 5), alpaca fibers generally have a non-circular cross-section, particularly the VOFs. Therefore, strictly speaking, alpaca fibers do not have a diameter, in the true sense of the word, since it does not have a circular shape. The term “fiber diameter” is generally defined as the distance of the vertical line between two points located on each edge of the longitudinal fiber image (IWTO 2015a; M. Quispe-Bonilla et al. 2022). Villarroel (1959) found ellipticity values between 1.15 and 1.23, and 1.18 and 1.28 for Suri and Huacaya alpaca fibers, respectively. Nevertheless, for fibers with major axes of 22.5 and 38.5 μm they found ellipticities of 1.10 and 1.24, respectively, which are lower than the values reported here. These differences may be due to their test method and the limited number of samples (only 5) they tested. Our ellipticity results are in line with those of B. A. McGregor and Quispe Peña (2018), namely between 1.29 and 1.34 for Huacaya alpaca fibers raised in Huancavelica – Perú.

Banky and Slen (1955) suggested that wool fibers must be considered as filaments of irregular cross section, while Atav and Hunter (2023) reported that coarse mohair fibers generally have a non-circular shape. The non-circularity of natural fibers was recently identified as a source of error when dealing with fiber tensile strength. Some authors have shown that the cross sections of natural fibers can be better represented by an ellipse rather than by a circle, especially for fibers of plant origin (Thomason 2023).

Perimeter information is very scarce, since the fiber perimeter is not widely measured or used, except for defining fiber shape, using the area/perimeter ratio (Allain and Thébault 1996), where a higher ratio indicates a more circular shape, and a lower ratio a more elliptical or flattened shape (non-circular). An index commonly used as a measure of circularity is the ratio of the perimeter squared (R^2) against the A, according the next formula: $R = \frac{P^2}{A}$, where R, P, and A are the circularity, perimeter, and area, respectively, with a value of $R = 4\pi$ indicating a perfectly round shape of the fiber cross section (Xu, Dong, and Chen 2007).

The differences between the cross-sectional characteristics of VNOF and VOF found here (Table 5 and Figure 7) are in line with the results of Villarroel (1959) and B. A. McGregor and Quispe Peña (2018) for alpaca fibers; those of Banky and Slen (1955) and Kicińska-Jakubowska et al. (2024) for wool fibers, and also those of Allain and Thébault (1996) and Atav and Hunter (2023) for mohair. These researchers generally reported that finer fibers have a lower area, perimeter, major and minor axes, and ellipticity than coarser fibers. Additionally, B. A. McGregor and Quispe Peña (2018) showed that ellipticity increases by 0.006 for each μm increase in the average fiber diameter.

The wide differences between the cross-sectional characteristics of VNOF and VOF could perhaps assist in identifying and quantifying the VOF in alpaca fibers using logistic regression and/or discriminant analysis, in white Huacaya alpacas. Thus, it would be possible to develop and implement a suitable software in an instrument to enable this to be done while measuring other cross-sectional characteristics. In this way, farmers, technicians, and other persons can use the device as a practical means to determine the incidence of VOF in alpaca fibers, and this could be extended to wool, mohair, llama, camel, and other animal fibers, since the presence and level of VOF represents a serious quality and commercial problem (Hunter, Smuts, and Botha 2013), while there is as yet, no way to accurately determine the incidence of this fiber type in a fast and practical way. Work aimed at measuring coarse and kemp fiber levels in mohair (Shakyawar et al. 2013) and other animal fibers (Allain and Thébault 1996) using digital imaging analysis techniques never attained technological or commercial success. Although the projection microscope represents a direct method of measuring and identifying coarse and medullated fibers, it is not suitable for practical every day use, due to the time, cost, and labor involved.

Medullation characteristics

With respect to the medullation and fiber diameter results (obtained with the Fiber Med instrument), it was found that VOF had a higher p_Med (more than 3 times), p_Cont_Med (more than 7 times), p_Str_Med (more than 480 times), and p_Disc_med (more than 1.7 times) than VNOF. Furthermore, the VNOFs had

a lower p_NoMed (more than 61 times) and a higher p_Frag_Med (more than 5 times) than VOFs. Moreover, the fiber diameters of VOFs with different medullation types are also greater than the corresponding VNOFs medullation types. Additionally, there is not an overlap between the AFD_{FM} ranges (maximum – minimum), of the VOFs and VNOFs (17.02–22.75 μm and 24.99–47.22 μm , respectively), the same applies to the AFD_{FM} of Med_Frag, Med_Discon, and Med_Cont fibers (see Table 6).

The differences between VNOF and VOF, in terms of fiber diameter (AFD) and medullation (p_Med), are related to the positive correlation between these two variables, as demonstrated by several researchers (Cruz et al. 2024; Pinares et al. 2018; E. Quispe et al. 2022) who reported the existence of a direct phenotypic and genetic relationship between them.

In addition, the medullation results obtained here on VOF snippets (Table 6) are very close to those obtained with the Medullometer (Table 4). The predominance of fibers with Cont_Med and Str_Med was 80.5% and 17.5% for the VOF whole fibers versus 74.16% and 14.41% for the VOF fiber snippets, respectively. Also, the results were similar for p_M (100% vs 98.9%) for the two methods (100% vs 98.9%). The presence of NoMed fibers in VOF snippets is due to the fact that no medullation was found in certain segments or sections of the whole VOF.

The AFD_{FM} ranges of VNOF (17.02–22.75 μm) and VOF (24.99–47.22 μm) do not overlap, hence this feature is also important in the identification and prediction of VOF in white Huacaya alpaca fibers using logistic regression or discriminant analysis. Furthermore, since the percentage of fibers coarser than 30 μm is a good predictor of the itching sensation associated with knitted fabrics, and even much more pronounced for woven fabrics, it is used as a comfort factor threshold (Hansford 1992; G. Naylor 1992). Nevertheless, the comfort threshold for alpaca fibers might be even lower, since our results show that there are VOFs even as fine as 25 μm which may produce itchiness. Moreover, Naebe et al. (2015) reported that much finer fibers (20 microns) may trigger the itching response if the free length of the protruding fiber from the surface of the fabric is short enough. For this reason, the comfort factor, normally defined as the percentage of fibers <30 μm , must be used with caution for alpaca fibers.

Table 6. Summary statistics for the medullation characteristics of non-objectionable and objectionable alpaca fibers ($n = 52$).

Characteristics	Mean	Standard deviation	Median	Minimum	Maximum	Standard error
<i>Non-objectionable fibers</i>						
p_Med (%)	30.24	18.19	29.3	2.2	72.14	2.52
SE_p_Med (%)	0.97	0.29	1.04	0.3	1.42	0.04
p_NoMed (%)	69.79	18.19	70.86	27.83	97.8	2.52
p_Frag_Med (%)	15.72	9.57	14.07	1.91	37.42	1.33
p_Disc_Med (%)	4.09	3.53	3.08	0.1	15.92	0.49
p_Cont_Med (%)	10.36	8.33	8.28	0.18	44.24	1.15
p_Str_Med (%)	0.03	0.03	0.03	0.00	0.13	0.00
AFD _{FM} (μm)	20.46	1.32	20.54	17.02	22.75	0.18
SD_AFD _{FM} (μm)	4.14	0.69	4.10	2.91	5.93	0.10
AFD_NoMed (μm)	19.23	0.98	19.31	16.6	20.92	0.14
AFD_Frag_Med (μm)	21.95	1.47	21.87	18.8	26.67	0.2
AFD_Disc_Med (μm)	22.63	1.55	22.81	18.75	25.75	0.21
AFD_Cont_Med (μm)	26.01	1.69	25.95	21.72	29.57	0.23
AFD_Str_Med (μm)	49.81	13.42	49.66	22.48	76.6	2.37
<i>Objectionable fibers</i>						
p_Med (%)	98.87	1.56	99.47	93.35	100.00	0.22
SE_p_Med (%)	0.64	0.70	0.43	0.00	2.88	0.10
p_NoMed (%)	1.13	1.56	0.52	0.00	6.65	0.22
p_Frag_Med (%)	3.25	3.81	2.29	0.00	20.67	0.53
p_Disc_Med (%)	7.05	7.24	5.24	0.00	38.56	1.00
p_Cont_Med (%)	74.16	10.62	75.04	27.31	90.61	1.47
p_Str_Med (%)	14.41	10.00	11.96	2.00	47.82	1.39
AFD _{FM} (μm)	33.03	4.66	32.10	24.99	47.22	0.65
SD_AFD _{FM} (μm)	11.70	2.40	11.45	7.33	17.19	0.33
AFD_NoMed (μm)	32.58	6.14	31.20	25.31	52.73	1.07
AFD_Frag_Med (μm)	31.14	3.85	30.73	24.42	42.16	0.59
AFD_Disc_Med (μm)	30.74	2.57	30.58	26.56	39.84	0.37
AFD_Cont_Med (μm)	35.41	3.13	35.07	29.13	45.83	0.43
AFD_Str_Med (μm)	50.35	5.78	49.33	36.91	66.79	0.81

p, indicates percentage; ADF, average fiber diameter; SE, standard error; SD, standard deviation; FM, indicates the Fiber Med device. Medulla types: p_Med, total medullation; NoMed, non-medullated; Frag, fragmented; Disc, discontinuous; Cont, continuous; Str, strong.

Table 7. Summary statistics for the fiber diameter characteristics of non-objectionable and objectionable fibers ($n = 52$).

Characteristics	Mean	Standard deviation	Median	Minimum	Maximum	Standard error
<i>Non-objectionable fibers</i>						
AFD _{FEC} (μm)	21.02	2.08	20.88	16.33	28.84	0.28
SD_AFD _{FEC} (μm)	4.42	0.61	4.44	3.32	6.16	0.08
Var_AFD _{FEC} (μm ²)	19.90	5.63	19.73	11.05	37.94	0.77
CV_AFD _{FEC} (%)	21.06	2.51	21.04	16.15	27.41	0.34
Fibers < 30 μm (%)	95.22	5.84	96.75	61.58	100.00	0.79
Fibers ≥ 30 μm (%)	4.78	5.84	3.24	0.00	38.42	0.79
Spinning fineness (μm)	20.49	1.97	20.34	15.98	27.50	0.27
<i>Objectionable fibers</i>						
AFD _{FEC} (μm)	32.80	4.80	31.31	24.10	45.09	0.46
SD_AFD _{FEC} (μm)	7.90	1.77	7.39	5.33	12.53	0.17
Var_AFD _{FEC} (μm ²)	65.48	30.51	54.67	28.39	157.02	2.90
CV_AFD _{FEC} (%)	24.25	4.88	23.77	16.25	37.45	0.46
Fibers < 30 μm (%)	40.43	20.48	43.76	1.44	85.85	1.94
Fibers ≥ 30 μm (%)	59.57	20.48	56.24	14.15	98.56	1.94
Spinning fineness (μm)	32.94	4.72	31.72	23.82	46.70	0.45

AFD, average fiber diameter; SD, standard deviation; Var, variance; CV, coefficient of variation.

The results for the average fiber diameter and its variability obtained with the Fiber EC device are given in Table 7, from which it can be seen that the AFD differences between VOF and VNOF are very similar to those in Table 6 (obtained with the Fiber Med instrument), with the variables, such as SD_AFD_{FEC}, CV_AFD_{FEC}, Fiber ≥ 30 μm, and Spinning fineness, also being higher, on average for the VOF than for the VNOF. Furthermore, it was found that more than 95% of VNOF are finer than 30 μm which is in contrast with VOF, where only some 40% are finer than 30 μm.

Our results for the diameter characteristics of VNOF generally fall within the ranges found by other researchers for un-dehaired alpaca (Montes et al. 2008; PECSA 2016; E. C. Quispe et al. 2009). The presence of coarse objectionable fibers in alpaca is associated with a slight increase (by 1.5 μm) in the AFD (Wang, Singh, and Wang 2008), while for llama, the difference between dehaired and un-dehaired fibers is more than 4 μm (E. C. Quispe, Chipa, and Pinares 2015). Also, our results for AFD, CV_AFD_{FEC}, and Fibers ≥ 30 μm of VOF fall within the ranges found by Wang, Singh, and Wang (2008), who reported values between 29.3 and 48.5 μm, 33% and 34%, and 41% and 83%, respectively, found for coarser alpaca after dehairing, using a cashmere dehairing machine.

Correlations

Correlations between fiber cross-sectional characteristics and AFD according to fiber medullation type for VNOF and VOF are shown in Table 8, in the form of a Pearson correlation matrix. The correlations between AFD_{FM} and the other cross-sectional characteristics (except ellipticity) are high ($r > 0.91$) and highly significant (p-value < .01) for VNOF. For VOF, the correlations are slightly lower, but still quite high ($r > 0.72$) and significant. The correlations, between ellipticity and AFD_{FM} and SD_AFD_{FM} are not significant for VNOF, whereas for VOF, they are moderate, positive, and significant ($r = 0.46$ and 0.49 , respectively). For VNOF, the correlations between SD_AFD_{FM} and the other cross-sectional characteristics are moderate ($r \approx 0.6$) and highly significant (p-value < .05), while, for VOF, they are slightly higher ($r > 0.86$) and also significant (p-value < .05).

In line with the above, only moderate correlations have been found between the cross-sectional characteristics (except ellipticity) and the AFD_Disc_Med and AFD_Frag_Med (r varies between 0.30 and 0.47), for both VNOF and VOF. With AFD_NoMed, on the other hand, they are slightly higher for VNOF (r varies about from 0.71 to 0.8), but are relatively low (r varies between 0.23 and 0.25) for VOF. For VNOF, the correlations between cross-sectional and AFD_Cont_Med, are moderate (r around 0.45), while for VOF, they are slightly higher (r varies around 0.71). Furthermore, a low, or non-significant correlation was found between the cross-sectional characteristics and AFD_Str_Med, for both VNOF and VOF.

Further to the above, it was found that, for VNOF, the correlations between AFD_{FM} and AFD_e and AFD_a are high ($r \approx 0.92$) and highly significant for VNOF, but are slightly lower for VOF ($r \approx 0.84$). In the case of the mean diameters of Frag_Med, Disc_Med and Cont_Med, the correlations are moderate

Table 8. Correlation between the various average fiber diameter and cross-sectional characteristics for non-objectionable and objectionable fibers, respectively.

Characteristics	AFD _{FM}	SD_ AFD _{FM}	AFD_ NoMed	AFD_ Frag_Med	AFD_ Disc_Med	AFD_ Cont_Med	AFD_ Str_Med
<i>Non-objectionable fibers (VNOF)</i>							
Area	0.928***	0.610***	0.745***	0.312*	0.337*	0.466***	−0.065 ^{ns}
Perimeter	0.926***	0.612***	0.754***	0.314*	0.328*	0.463***	−0.088 ^{ns}
Minor axis	0.910***	0.558***	0.713***	0.303*	0.309*	0.434**	−0.092 ^{ns}
Mayor axis	0.908***	0.631***	0.760***	0.312*	0.327*	0.468***	0.090 ^{ns}
Ellipticity	−0.111 ^{ns}	0.096 ^{ns}	0.035 ^{ns}	−0.023 ^{ns}	0.001 ^{ns}	0.014 ^{ns}	0.028 ^{ns}
AFD_e	0.926***	0.603***	0.748***	0.314*	0.329*	0.461***	−0.081 ^{ns}
AFD_a	0.926***	0.610***	0.753***	0.314*	0.325*	0.461***	0.092 ^{ns}
AFD _{FM}		0.666***	0.817***	0.390**	0.468***	0.620***	−0.037 ^{ns}
SD_AFD _{FM}			0.254 ^{ns}	−0.060 ^{ns}	0.045 ^{ns}	0.491***	0.167 ^{ns}
AFD_NoMed				0.708***	0.684***	0.571***	−0.021 ^{ns}
AFD_Frag_Med					0.821***	0.523***	0.061 ^{ns}
AFD_Disc_Med						0.705***	0.083 ^{ns}
AFD_Cont_Med							0.171 ^{ns}
<i>Objectionable fibers (VOF)</i>							
Area	0.826***	0.861***	0.245 ^{ns}	0.277 ^{ns}	0.516***	0.724***	0.263 ^{ns}
Perimeter	0.845***	0.886***	0.249 ^{ns}	0.283 ^{ns}	0.512***	0.721***	0.278*
Minor axis	0.722***	0.749***	0.259 ^{ns}	0.163 ^{ns}	0.508***	0.707***	0.235 ^{ns}
Mayor axis	0.848***	0.893***	0.230 ^{ns}	0.311*	0.463***	0.679***	0.278*
Ellipticity	0.457***	0.490***	0.039 ^{ns}	0.280 ^{ns}	0.020 ^{ns}	0.193 ^{ns}	0.155 ^{ns}
AFD_e	0.835***	0.871***	0.256 ^{ns}	0.267 ^{ns}	0.528***	0.739***	0.280*
AFD_a	0.842***	0.883***	0.249 ^{ns}	0.278 ^{ns}	0.515***	0.723***	0.276 ^{ns}
AFD _{FM}		0.885***	0.253 ^{ns}	0.164 ^{ns}	0.534***	0.886***	0.494***
SD_AFD _{FM}			0.267 ^{ns}	0.255 ^{ns}	0.492***	0.717***	0.429**
AFD_NoMed				0.252 ^{ns}	0.268 ^{ns}	0.032 ^{ns}	0.181 ^{ns}
AFD_Frag_Med					0.019 ^{ns}	0.076 ^{ns}	0.071 ^{ns}
AFD_Disc_Med						0.597***	0.346*
AFD_Cont_Med							0.421**

ADF, average fiber diameter; SD, standard deviation; FM, Fiber Med device. Medulla types: NoMed, non-medullated; Frag, fragmented; Disc, discontinuous; Cont, continuous; Str, strong. AFD_e, average equivalent fiber diameter; AFD_a, average apparent fiber diameter. * Significance of the correlation ($p < .05$); ^{ns} not significant.

(r varies be from about 0.31 to 0.46) for VNOF, but are slightly lower for VOF. Nevertheless, for AFD_Cont_Med and AFD_Disc_Med, the correlations are much higher ($r \approx 0.52$ and $r \approx 0.74$, respectively).

The correlations between AFD_{FM} and AFD_Frag_Med and AFD_Disc_Med, AFD_Cont_Med for VNOF and VOF, respectively, obtained here are slightly different from those reported by E. Quispe et al. (2022) for alpaca, probably due to the fact that they tested whole fibers sample from which, VOF had not been removed. Furthermore, the high correlations obtained here between AFD_{FM} and AFD_e and AFD_a suggest that these three features may be good indicators of fiber cross-sectional size (thickness). “AFD” on its own is not entirely accurate due to the alpaca fiber not being circular in shape. Furthermore, the area, perimeter, and minor and major axes are also good fiber “thickness” indicators (Sommerville 1998) due to the sectional way in which the fibers have been measured, and also due to their high correlations with the different types of fiber diameter measurements. Furthermore, according to the present results, AFD_{FM} can also be used to calculate the fiber cross-sectional area and perimeter, assuming that the cross-section of the fiber is round (Peng et al. 2018). The other correlations, between cross-sectional characteristics and AFD, for the different medullation types are new.

With respect to the correlations between the cross-sectional characteristics and the percentages of different medullation types (p_NoMed, p_Frag_Med, p_Disc_Med, p_Cont_med, and p_Str_Med), as well as the different measures of mean fiber diameter (AFD_{FM}, AFD_e, and AFD_a) for VNOF and VOF (See Table 9), it was found that they vary from moderate to high ($r \approx [0.43-0.68]$). The correlations between the cross-sectional characteristics and p_NoMed are negative and fairly high ($r \approx -0.7$) for VNOF. For VOF, these correlations are moderate and negative, except for p_Str_Med, where they are not significant for VNOF, but are significant ($p < .05$) and high ($r \approx [0.54 - 0.83]$) for VOF. Furthermore, it was found that p_NoMed had highly significant ($p < .05$) and negative correlations with p_Frag_Med ($r = -0.88$), p_Disc_Med ($r = -0.91$), and p_Cont_Med ($r = -0.78$), but lower correlations with p_Str_Med ($r = -0.31$) for VNOF. In contrast to this, for VOF these correlations are positive with p_Frag_Med and p_Disc_Med, but negative with p_Cont_Med and p_Str_Med.

Table 9. Correlation between fiber cross-sectional of characteristics (measured with fiber Mult) and fiber medullation (measured with fiber Med).

Characteristics	p_Med	SE_p_Med	p_NoMed	p_Frag_Med	p_Disc_Med	p_Cont_Med	p_Str_Med
<i>Non-objectionable fibers (VNOF)</i>							
Area	0.679***	0.713***	−0.679***	0.661***	0.592***	0.474***	0.145 ^{ns}
Perimeter	0.662***	0.704***	−0.662***	0.644***	0.569***	0.465***	0.134 ^{ns}
Minor axis	0.702***	0.736***	−0.702***	0.683***	0.622***	0.485***	0.113 ^{ns}
Major axis	0.601***	0.658***	−0.610***	0.591***	0.509***	0.437***	0.179 ^{ns}
Ellipticity	−0.305*	−0.271 ^{ns}	0.305*	−0.298*	−0.346*	−0.178 ^{ns}	0.144 ^{ns}
AFD_e	0.674***	0.713***	−0.674***	0.657***	0.584***	0.469***	0.145 ^{ns}
AFD_a	0.664***	0.707***	−0.664***	0.645***	0.571***	0.468***	0.152 ^{ns}
p_Med		0.849***	−1.000***	0.883***	0.913***	0.782***	0.305*
SE_p_Med			−0.849***	0.828***	0.711***	0.602***	0.273 ^{ns}
p_NoMed				−0.883***	−0.913***	−0.782***	−0.305*
p_Frag_Med					0.876***	0.409**	0.120 ^{ns}
p_Disc_Med						0.562***	0.226 ^{ns}
p_Cont_Med							0.428**
<i>Objectionable fibers (VOF)</i>							
Area	0.342*	−0.353**	−0.342*	−0.535***	−0.542***	−0.121 ^{ns}	0.737***
Perimeter	0.343*	−0.361**	−0.343*	−0.536***	−0.553***	−0.143 ^{ns}	0.768***
Minor axis	0.315*	−0.321*	−0.315*	−0.541***	−0.481***	0.031 ^{ns}	0.540***
Major axis	0.331*	−0.354**	−0.331*	−0.497***	−0.550***	−0.219 ^{ns}	0.827***
Ellipticity	0.129 ^{ns}	−0.155 ^{ns}	−0.129 ^{ns}	−0.098 ^{ns}	−0.278*	−0.387**	0.638***
AFD_e	0.343*	−0.358**	−0.343*	−0.546***	−0.544***	−0.099 ^{ns}	0.721***
AFD_a	0.341*	−0.369**	−0.341*	−0.538***	−0.551***	−0.135 ^{ns}	0.759***
p_Med		−0.892***	−1.000***	−0.730***	−0.460***	0.349**	0.372**
SE_p_Med			0.892***	0.644***	0.379**	−0.267 ^{ns}	−0.353**
p_NoMed				0.730***	0.460***	−0.349**	−0.372**
p_Frag_Med					0.764***	−0.521***	−0.463***
p_Disc_Med						−0.513***	−0.508***
p_Cont_Med							−0.424**

p, indicates percentage; ADF, average fiber diameter; SE, standard error. Medulla types: NoMed, non-medullated; Frag, fragmented; Disc, discontinuous; Cont, continuous; Str, strong. AFD_e, average equivalent fiber diameter; AFD_a, average apparent fiber diameter. *Significance of the correlation ($p < .05$); ^{ns} not significant.

According to Table 9, the correlations between AFD_e and AFD_a with p_Med are fairly high ($r \approx 0.67$) for VNOF, but are lower ($r \approx 0.34$) for VOF. In addition, the correlations between these two different measures of fiber diameter and the percentages of the different medullation types for VNOF are positive and moderate, except for p_Str_Med, which is not significant. For VOF, on the other hand, these correlations are negative, except for p_Str_Med, which is positive and high (r varies between 0.72 and 0.76).

Our correlations between AFD_e and AFD_a and p_Disc_Med, p_Cont_Med, p_Str_Med, and p_NoMed for VNOF are fairly similar to those between AFD and p_Disc_Med, p_Cont_Med, p_Str_Med, and p_NoMed found by Olarte et al. (2023) for alpaca (0.58, 0.47, 0.15, and −0.67, respectively). The correlations with p_Frag_Med are, however, different (−0.01 vs −0.67), probably due to the different methodologies and the origin of the animals. Also, the correlations for VOFs are very different, due to the differences in fiber types considered. The other correlations, namely between the cross-sectional characteristics and the percentages of the different medullation types, are new.

Figure 8 shows the correlations between the various cross-sectional characteristics and the different average fiber diameters for VNOF and VOF, respectively. It can be seen that the correlations between them – in general (combined) and separately for VNOF and VOF – vary from moderate to high ($r \approx [0.66–0.99]$), except for VNOF ellipticity. Likewise, these correlations are slightly different for VNOF and VOF. Thus, very high and positive correlations ($r \approx [0.95–0.99]$) were found between the different types of fiber diameter measurements (AFD, AFD_e, AFD_a), area, perimeter, and major and minor axes, although, in almost all cases, slightly lower values were found for VOF than for VNOF. Likewise, overall (i.e. VNOF and VOF combined) moderate to high correlations were found between these characteristics and ellipticity ($r \approx [0.66–0.81]$), although, between ellipticity and fibers $< 30 \mu\text{m}$, the correlation was negative ($r = -0.68$), whereas, when VNOF and VOF are considered separately, these correlations are very low and, in most cases, not significant ($p > .05$).

Geometrically, a circular shape implies that, for each diameter, a perimeter (circumference) and area are defined, with a perfect relationship between them ($r = 1$) (M. Sánchez and Molina 2013). However, when the shape is not circular, the relationship is not perfect, as is the case for our

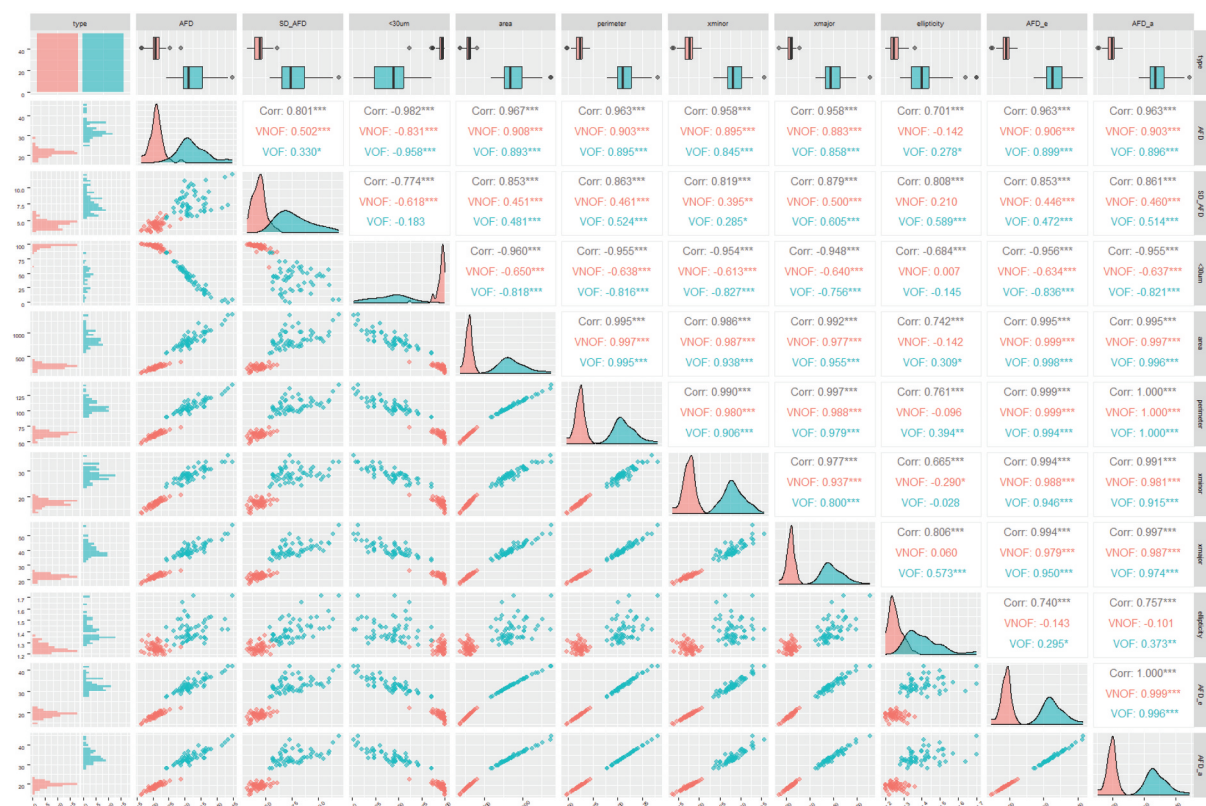


Figure 8. Chart correlation between cross-sectional and fiber diameter characteristics for visually non-objectionable fibers (VNOF) and visually objectionable fibers (VOF) in alpaca .

results of the inter-relationships between the various cross-sectional characteristics. This confirms once again that alpaca fibers do not have a perfectly circular or round shape. Hence, the correlations between those cross-sectional characteristics are greater for VNOF than for VOFs because VOFs are less circular than VNOF, as demonstrated by their ellipticity values (1.25 vs 1.41) and corresponding images (Figure 7). The results for the inter-relationships between cross-sectional characteristics for VOF and VNOF are new, but are consistent with geometrical theory.

Conclusions

This study has characterized visually objectionable and non-objectionable Huacaya alpaca fibers in terms of various dimensional, medullation, and diameter properties. VOFs were found to represent less than 2.5% of the sample by weight, with both monotypic and heterotypic fibers being present. All cross-sectional characteristics, incidence of medullation, mean fiber diameter, standard deviation of fiber diameter, percentage of fibers $\geq 30 \mu\text{m}$, and spinning fineness of the VOFs were higher than those of the visually non-objectionable ones. Except for ellipticity, all the correlations between the average fiber diameter and cross-sectional characteristics were high for VNOFs, but slightly lower for VOFs. The correlations between the cross-sectional characteristics and medullation levels were moderate for VOFs, but slightly lower for VOFs, except for ellipticity. Many of the results obtained in this study are novel and could be useful for identifying and quantifying VOFs, but this would be limited to white Huacaya alpacas only. Nevertheless, it is still necessary to develop appropriate technologies that would enable the rapid and accurate identification and quantification of such visually objectionable medullated fibers in practice, so that their incidence and levels can be reduced in practice through genetic selection and other means.

Authors' contributions

All authors contributed to Conceptualization, Methodology, Writing – original draft, Writing – review and editing.

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No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this research are available on request from the corresponding author.

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